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(54) **CAMERA AND LENS APPARATUS**

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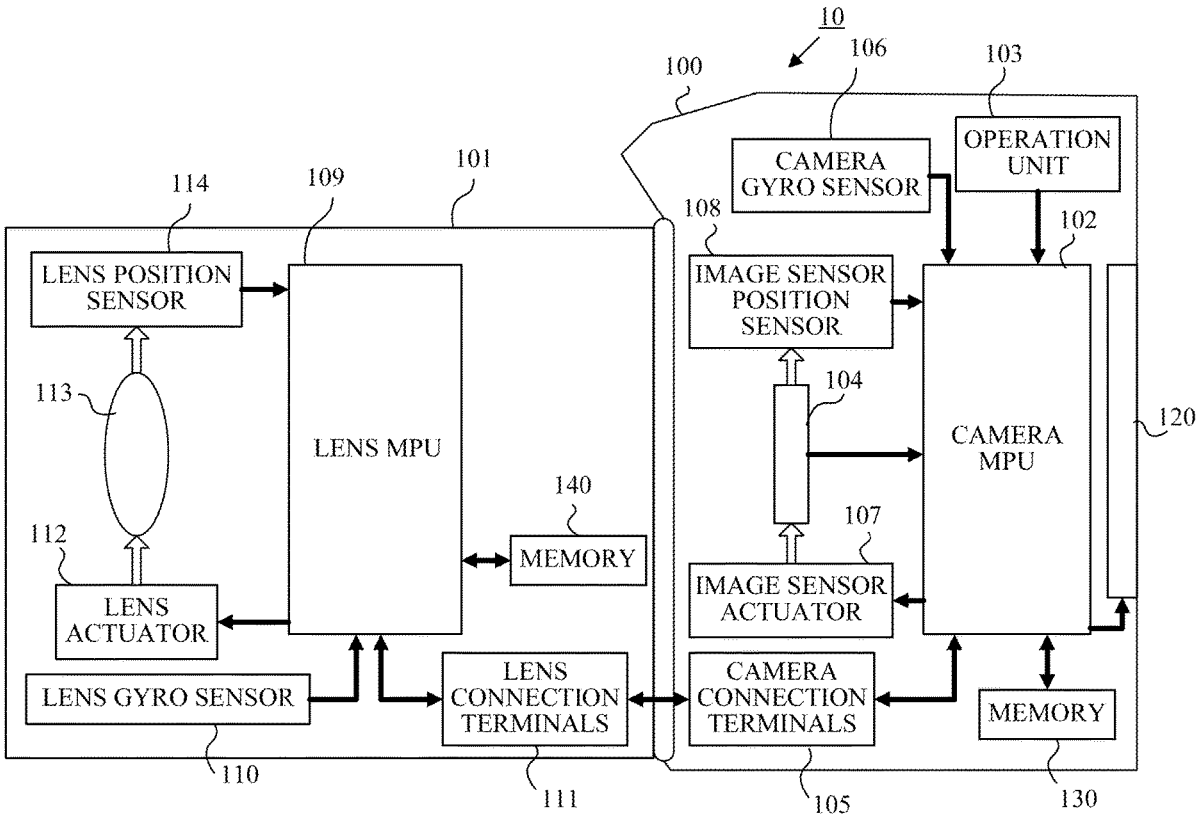
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(57) **ABSTRACT**

The camera is used in a camera system including the camera that includes an image sensor movable for image blur correction, and a lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction. The camera includes a setting unit configured to set a ratio of image blur correction amounts provided respectively by movements of the optical element and the image sensor in each of their moving directions, using information on an image circle formed by the lens apparatus, information on a mechanical movable amount of the image sensor, and information on a size of a signal readout area of the image sensor.



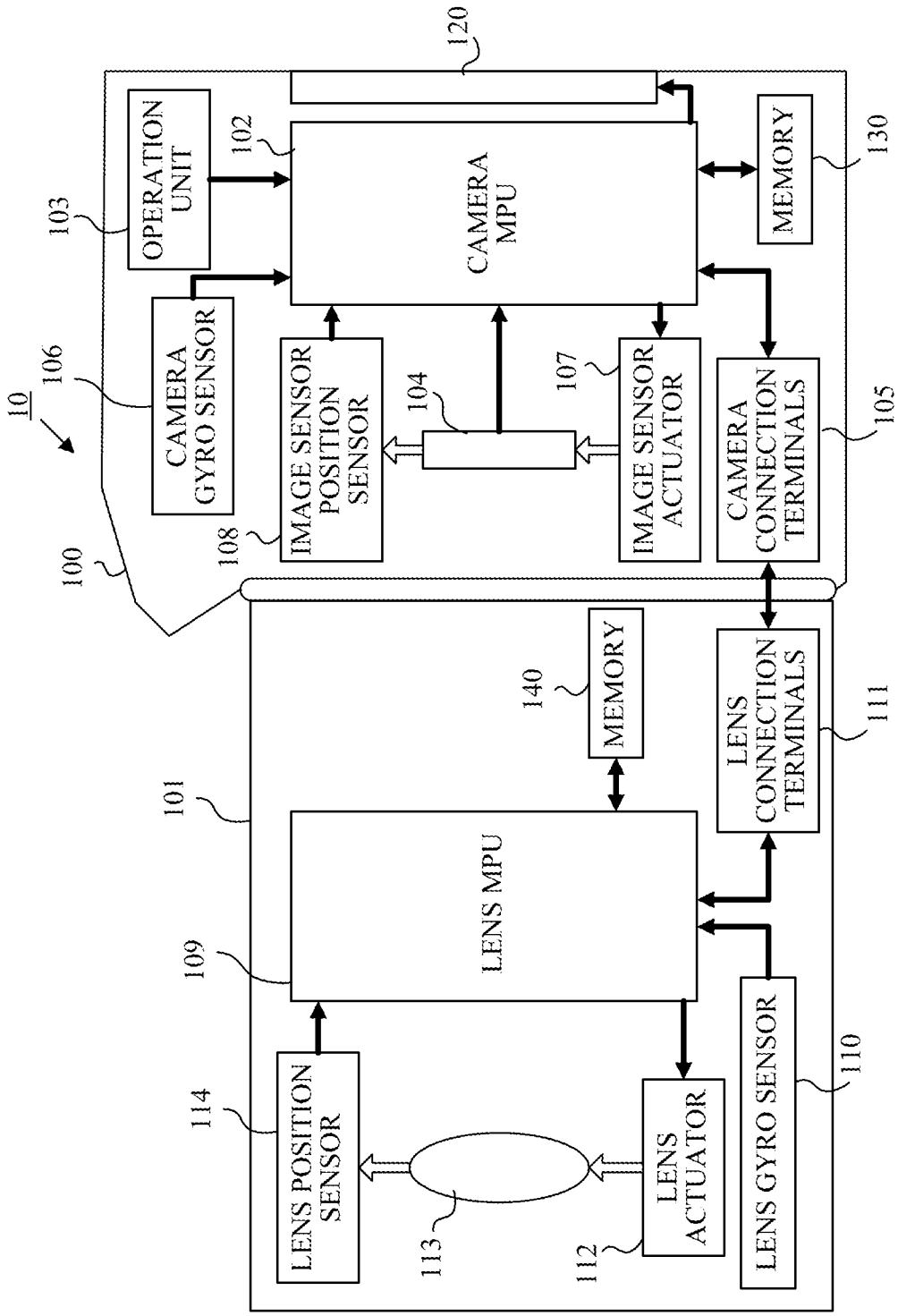

10
100
101
102
103
104
105
106
107
108
109
110
111
112
113
114
120
130
140
OPERATION UNIT
CAMERA MPU
MEMORY
CAMERA GYRO SENSOR
IMAGE SENSOR POSITION SENSOR
IMAGE SENSOR ACTUATOR
CAMERA CONNECTION TERMINALS
MEMORY
LENS CONNECTION TERMINALS
LENS MPU
LENS POSITION SENSOR
LENS ACTUATOR
LENS GYRO SENSOR

FIG. 1

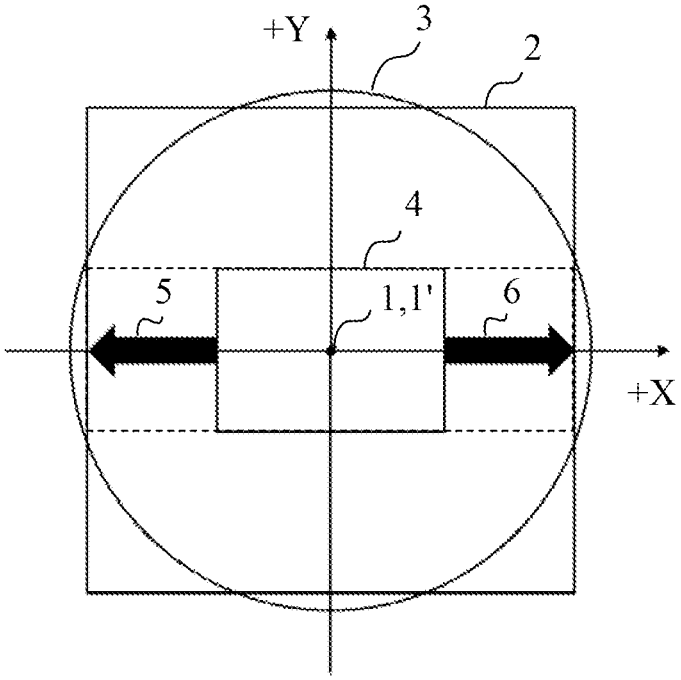


FIG. 2

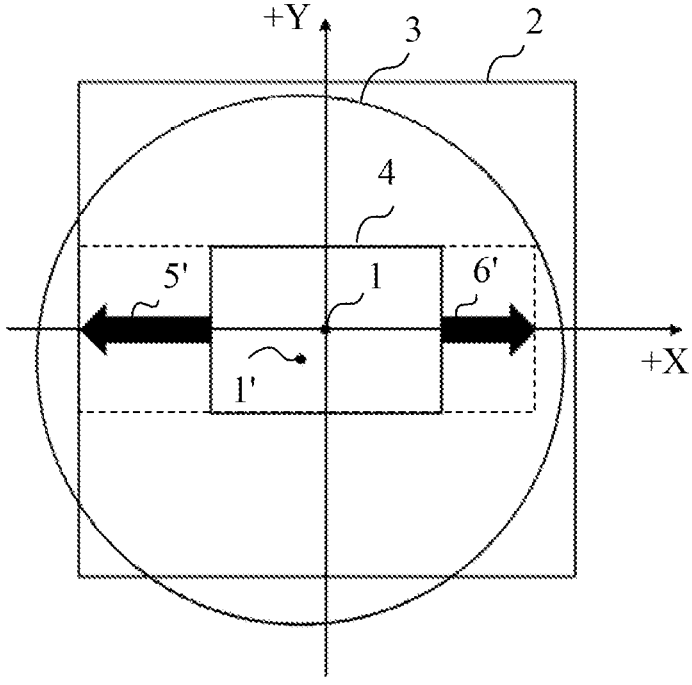


FIG. 3

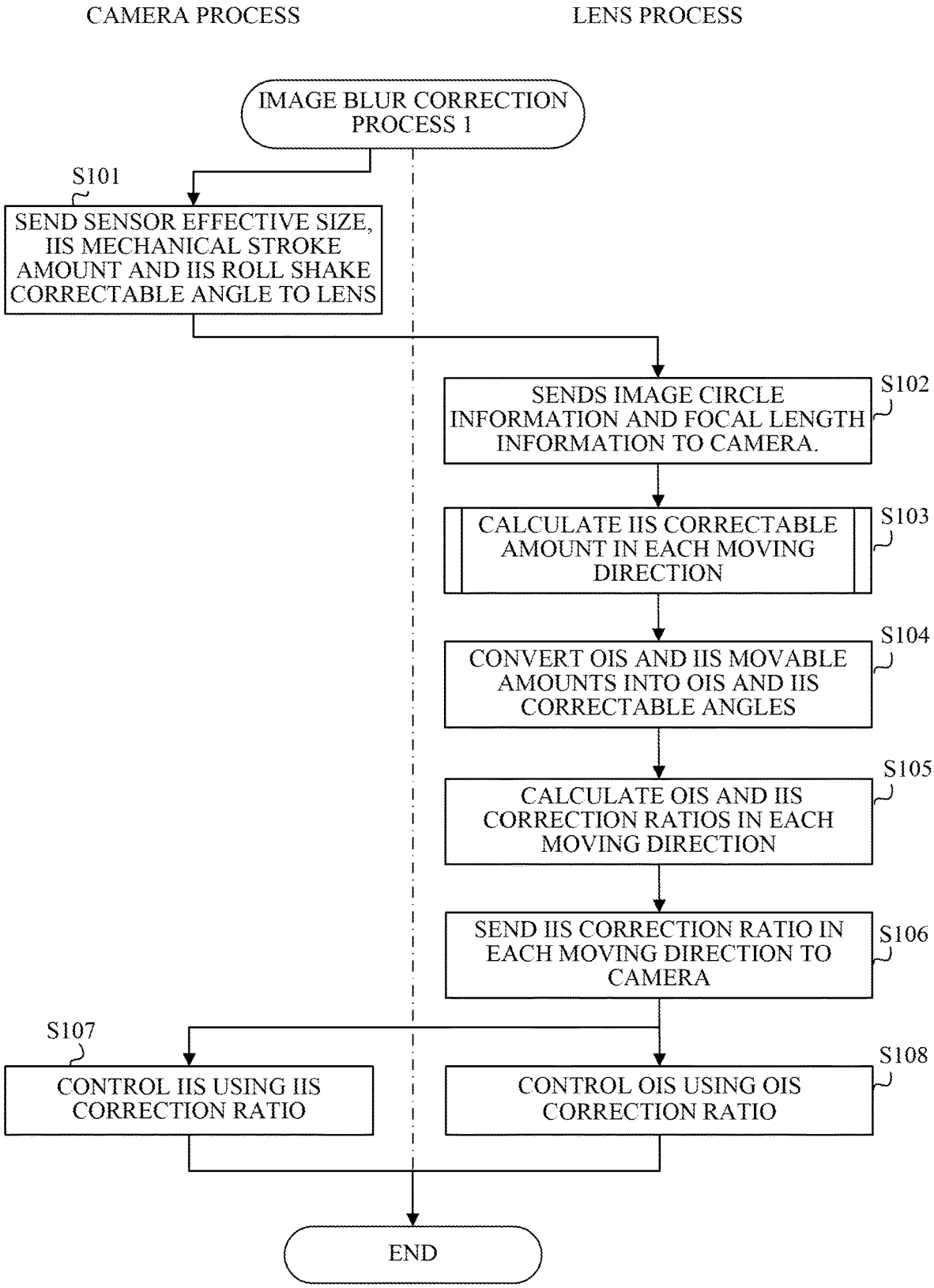

CAMERA PROCESS
LENS PROCESS
IMAGE BLUR CORRECTION PROCESS 1
S101
SEND SENSOR EFFECTIVE SIZE, IIS MECHANICAL STROKE AMOUNT AND IIS ROLL SHAKE CORRECTABLE ANGLE TO LENS
SENDS IMAGE CIRCLE INFORMATION AND FOCAL LENGTH INFORMATION TO CAMERA.
S102
CALCULATE IIS CORRECTABLE AMOUNT IN EACH MOVING DIRECTION
S103
CONVERT OIS AND IIS MOVABLE AMOUNTS INTO OIS AND IIS CORRECTABLE ANGLES
S104
CALCULATE OIS AND IIS CORRECTION RATIOS IN EACH MOVING DIRECTION
S105
SEND IIS CORRECTION RATIO IN EACH MOVING DIRECTION TO CAMERA
S106
S107
CONTROL IIS USING IIS CORRECTION RATIO
CONTROL OIS USING OIS CORRECTION RATIO
S108
END

FIG. 4

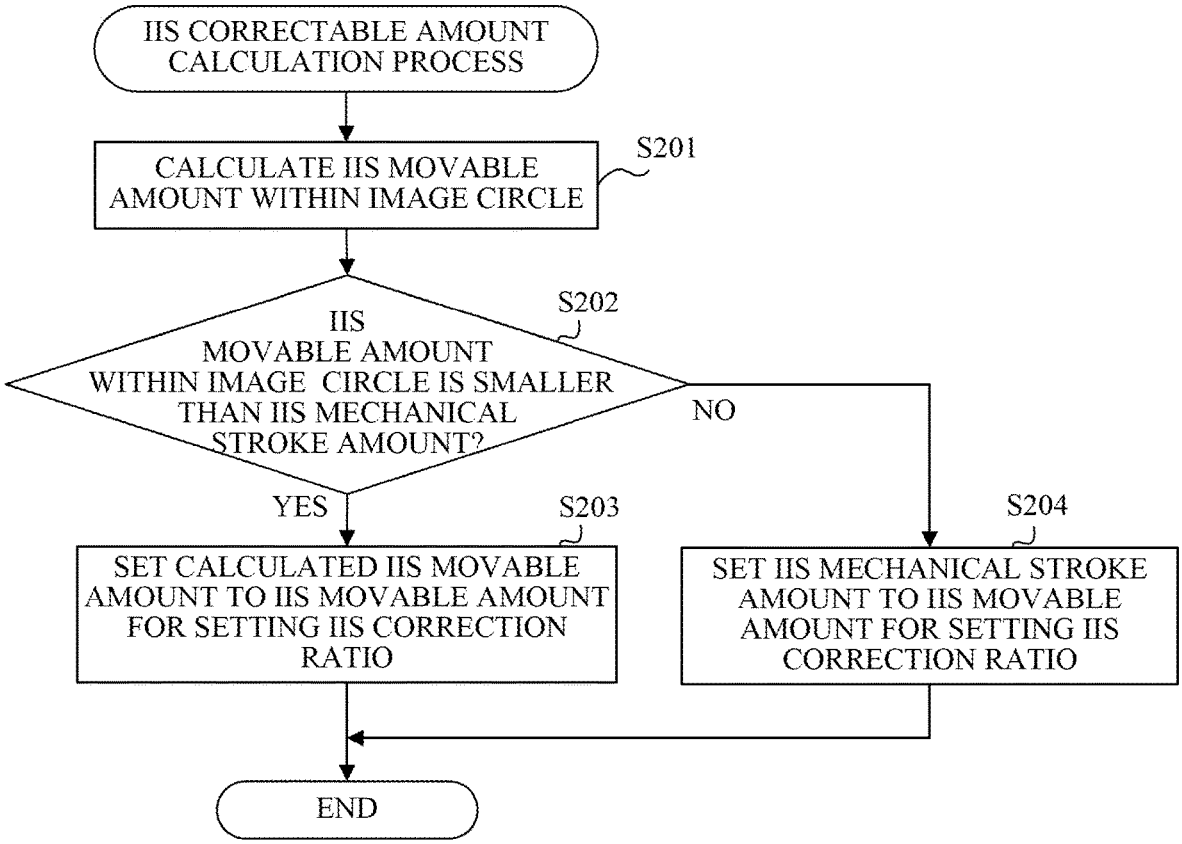

IIS CORRECTABLE AMOUNT CALCULATION PROCESS
CALCULATE IIS MOVABLE AMOUNT WITHIN IMAGE CIRCLE
S201
IIS MOVABLE AMOUNT WITHIN IMAGE CIRCLE IS SMALLER THAN IIS MECHANICAL STROKE AMOUNT?
S202
NO
YES
S203
SET CALCULATED IIS MOVABLE AMOUNT TO IIS MOVABLE AMOUNT FOR SETTING IIS CORRECTION RATIO
S204
SET IIS MECHANICAL STROKE AMOUNT TO IIS MOVABLE AMOUNT FOR SETTING IIS CORRECTION RATIO
END

FIG. 5

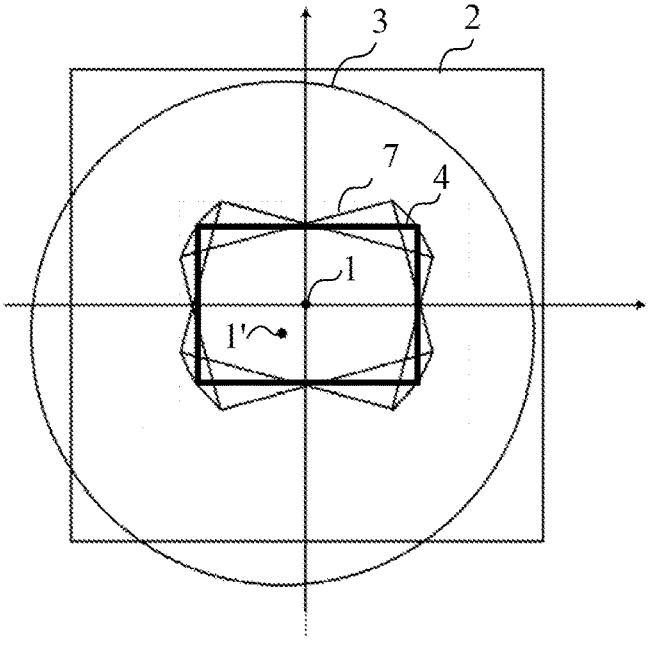

3
2
7
4
1
1'

FIG. 6

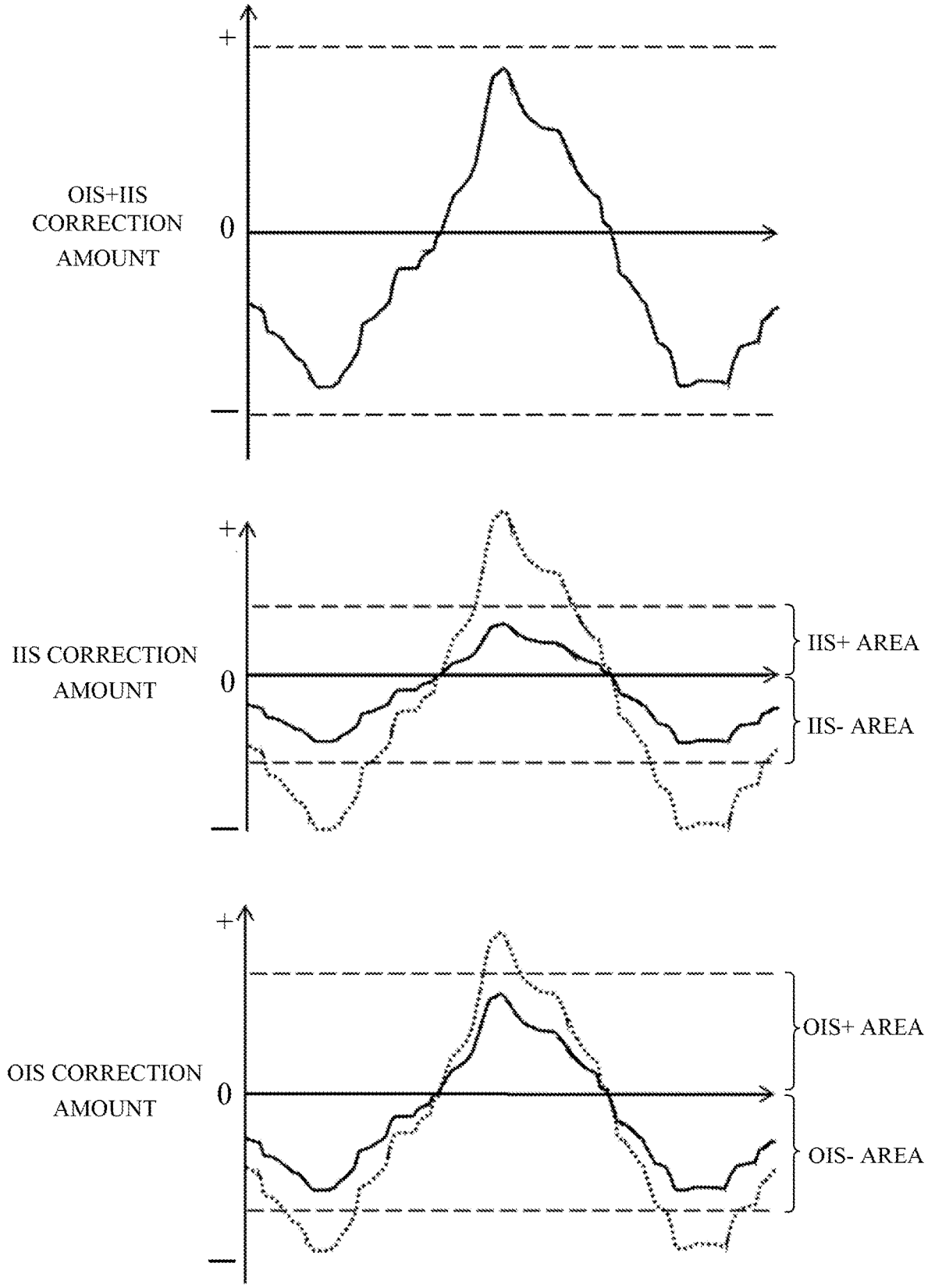

OIS+IIS
CORRECTION
AMOUNT
+
0
—
IIS CORRECTION
AMOUNT
IIS+ AREA
IIS- AREA
OIS CORRECTION
AMOUNT
OIS+ AREA
OIS- AREA

FIG. 7

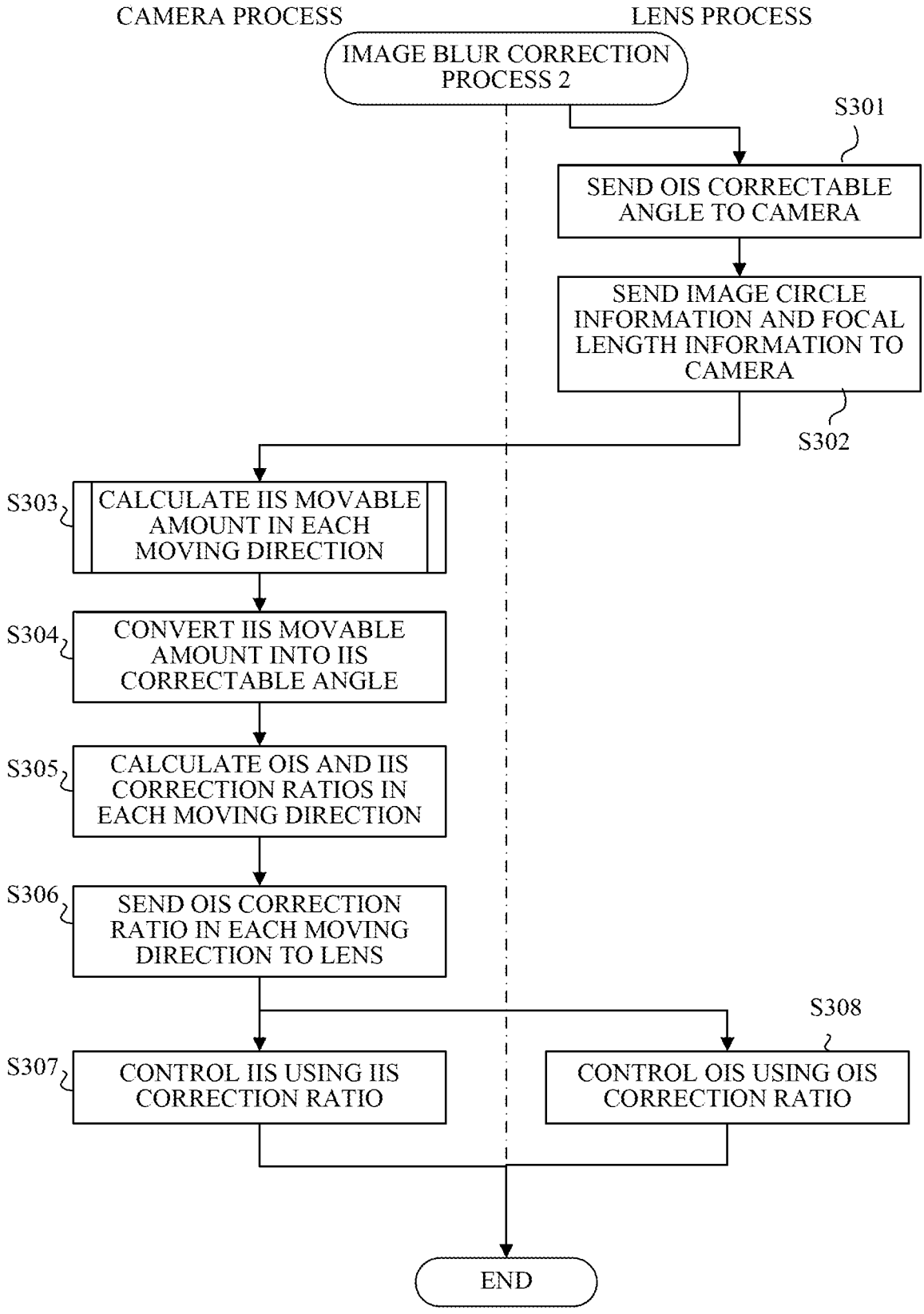

CAMERA PROCESS
LENS PROCESS
IMAGE BLUR CORRECTION PROCESS 2
S301
SEND OIS CORRECTABLE ANGLE TO CAMERA
SEND IMAGE CIRCLE INFORMATION AND FOCAL LENGTH INFORMATION TO CAMERA
S302
S303
CALCULATE IIS MOVABLE AMOUNT IN EACH MOVING DIRECTION
S304
CONVERT IIS MOVABLE AMOUNT INTO IIS CORRECTABLE ANGLE
S305
CALCULATE OIS AND IIS CORRECTION RATIOS IN EACH MOVING DIRECTION
S306
SEND OIS CORRECTION RATIO IN EACH MOVING DIRECTION TO LENS
S308
S307
CONTROL IIS USING IIS CORRECTION RATIO
CONTROL OIS USING OIS CORRECTION RATIO
END

FIG. 8

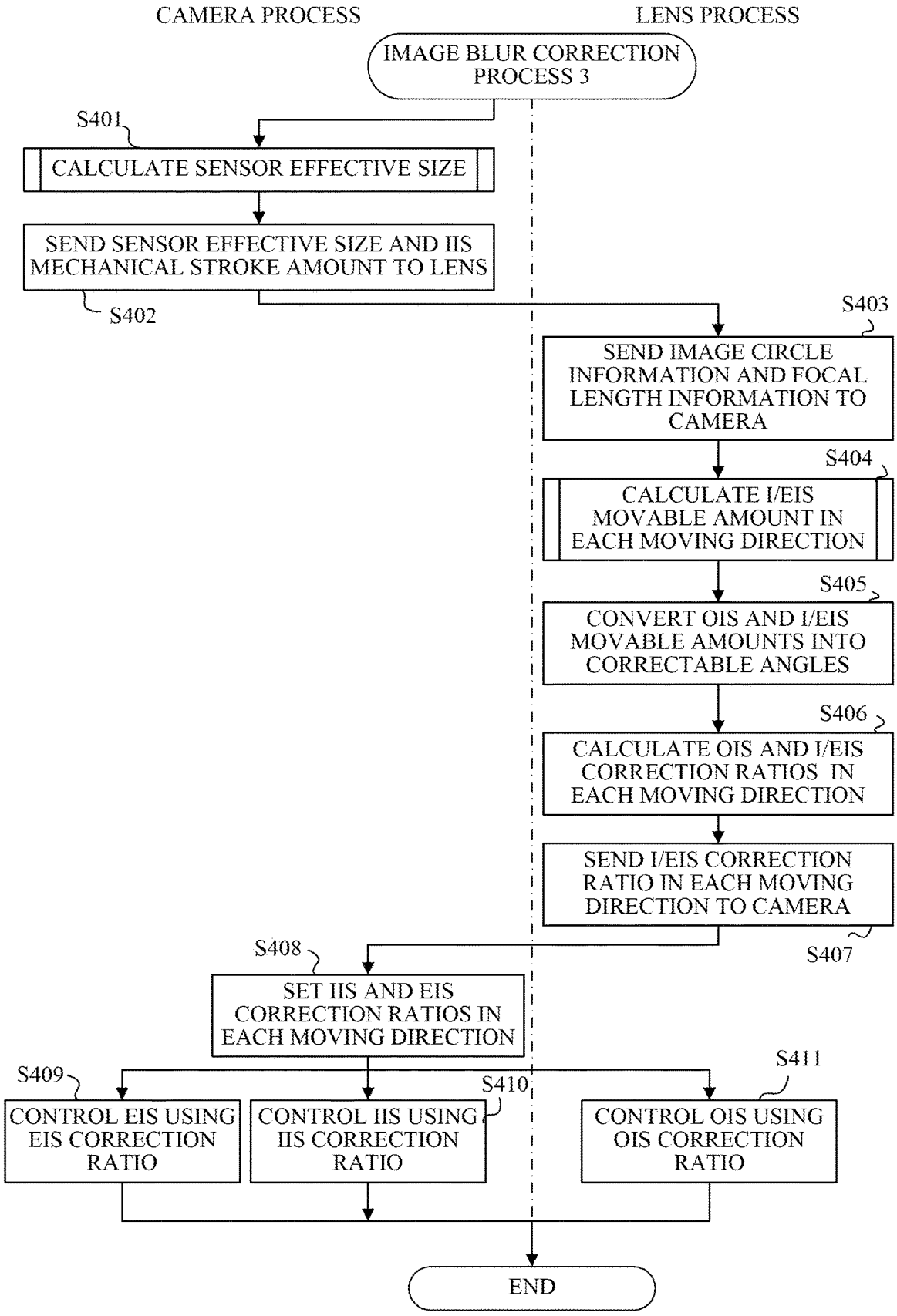

CAMERA PROCESS
LENS PROCESS
IMAGE BLUR CORRECTION PROCESS 3
S401
CALCULATE SENSOR EFFECTIVE SIZE
SEND SENSOR EFFECTIVE SIZE AND IIS MECHANICAL STROKE AMOUNT TO LENS
S402
S403
SEND IMAGE CIRCLE INFORMATION AND FOCAL LENGTH INFORMATION TO CAMERA
S404
CALCULATE I/EIS MOVABLE AMOUNT IN EACH MOVING DIRECTION
S405
CONVERT OIS AND I/EIS MOVABLE AMOUNTS INTO CORRECTABLE ANGLES
S406
CALCULATE OIS AND I/EIS CORRECTION RATIOS IN EACH MOVING DIRECTION
SEND I/EIS CORRECTION RATIO IN EACH MOVING DIRECTION TO CAMERA
S407
S408
SET IIS AND EIS CORRECTION RATIOS IN EACH MOVING DIRECTION
S409
CONTROL EIS USING EIS CORRECTION RATIO
S410
CONTROL IIS USING IIS CORRECTION RATIO
S411
CONTROL OIS USING OIS CORRECTION RATIO
END

FIG. 9

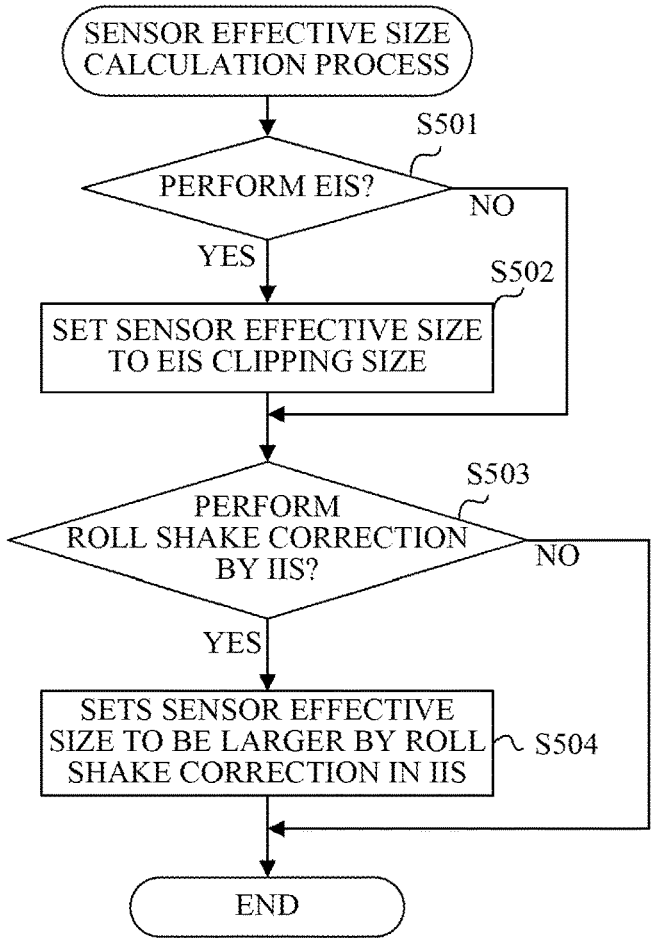

SENSOR EFFECTIVE SIZE CALCULATION PROCESS
S501
PERFORM EIS?
NO
YES
S502
SET SENSOR EFFECTIVE SIZE TO EIS CLIPPING SIZE
S503
PERFORM ROLL SHAKE CORRECTION BY IIS?
NO
YES
SETS SENSOR EFFECTIVE SIZE TO BE LARGER BY ROLL SHAKE CORRECTION IN IIS
S504
END

FIG. 10

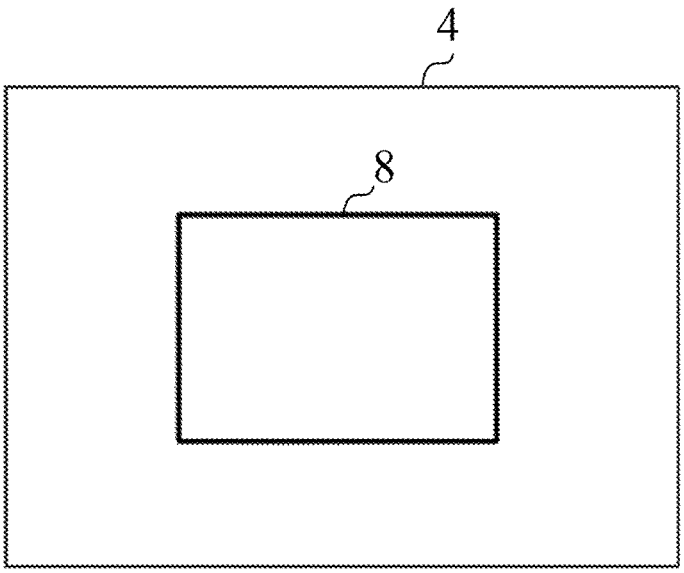

4
8

FIG. 11

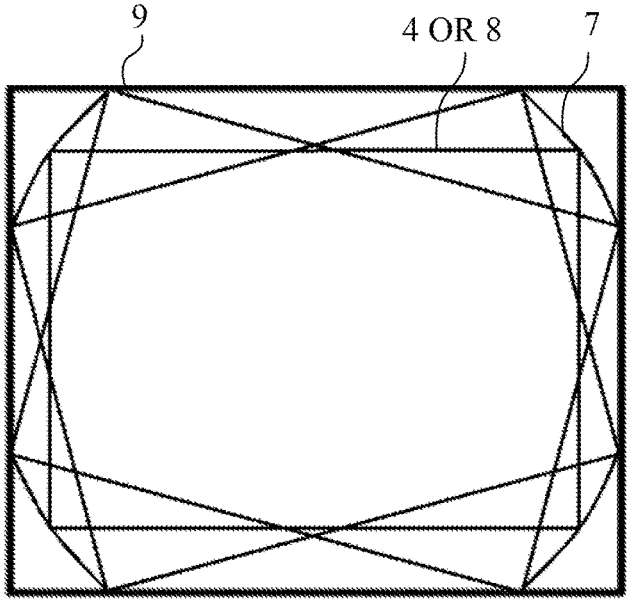


FIG. 12

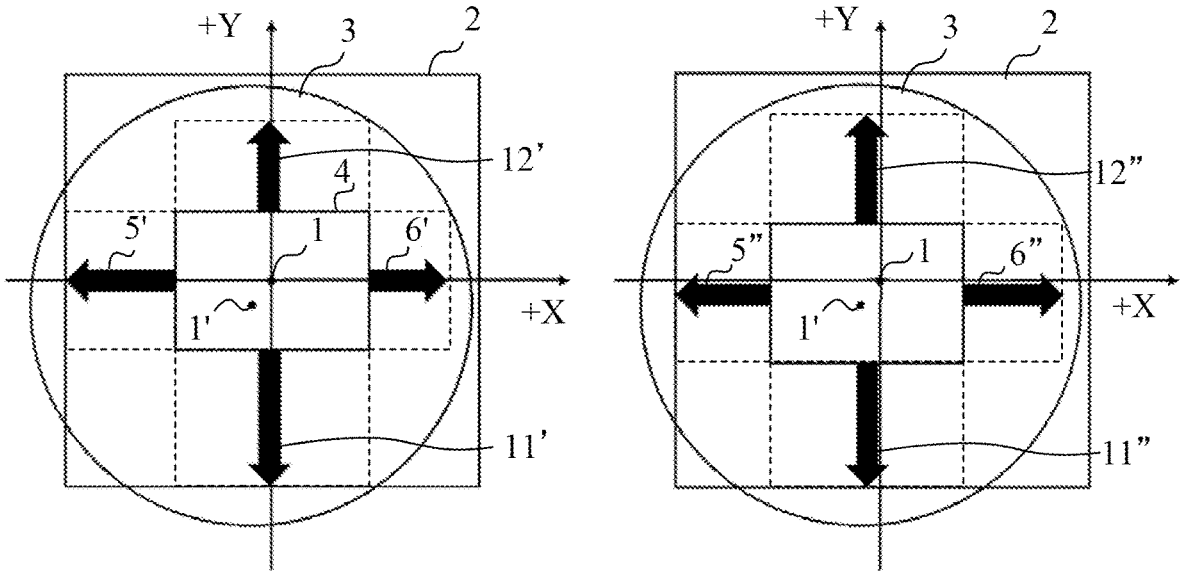


FIG. 13

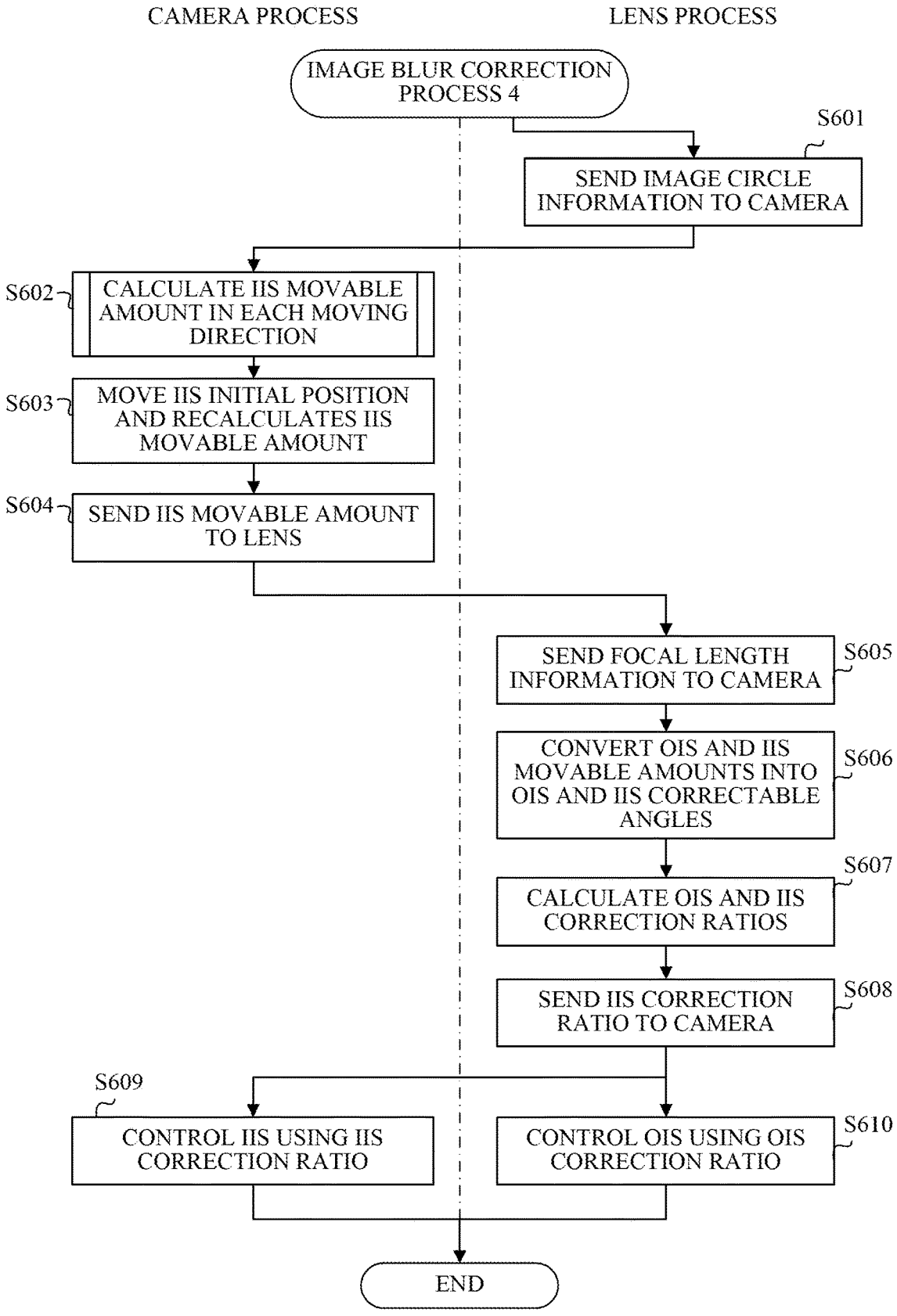

CAMERA PROCESS
LENS PROCESS
IMAGE BLUR CORRECTION PROCESS 4
S601
SEND IMAGE CIRCLE INFORMATION TO CAMERA
S602
CALCULATE IIS MOVABLE AMOUNT IN EACH MOVING DIRECTION
S603
MOVE IIS INITIAL POSITION AND RECALCULATES IIS MOVABLE AMOUNT
S604
SEND IIS MOVABLE AMOUNT TO LENS
SEND FOCAL LENGTH INFORMATION TO CAMERA
S605
CONVERT OIS AND IIS MOVABLE AMOUNTS INTO OIS AND IIS CORRECTABLE ANGLES
S606
CALCULATE OIS AND IIS CORRECTION RATIOS
S607
SEND IIS CORRECTION RATIO TO CAMERA
S608
S609
CONTROL IIS USING IIS CORRECTION RATIO
CONTROL OIS USING OIS CORRECTION RATIO
S610
END

FIG. 14

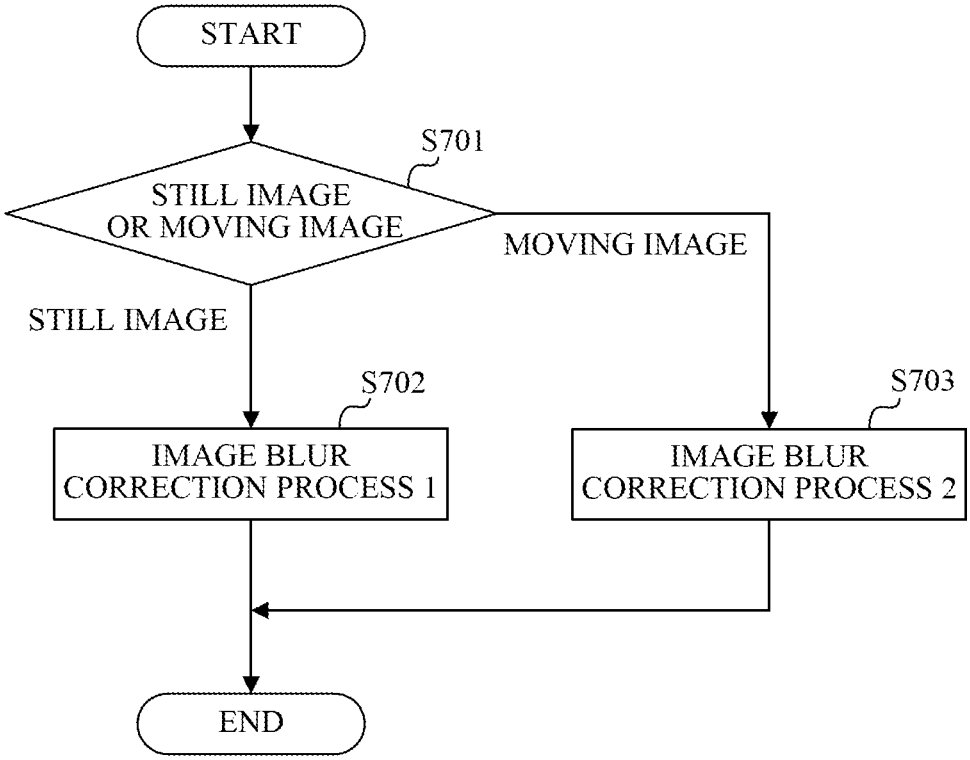

START
S701
STILL IMAGE
OR MOVING IMAGE
MOVING IMAGE
STILL IMAGE
S702
S703
IMAGE BLUR
CORRECTION PROCESS 1
IMAGE BLUR
CORRECTION PROCESS 2
END

FIG. 15

CAMERA AND LENS APPARATUS

BACKGROUND OF THE INVENTION

Field of the Invention

[0001] The present invention relates to optical apparatuses, that is, lens apparatuses and cameras used in camera systems having an image blur correction function.

Description of the Related Art

[0002] Lens-interchangeable camera system having an image blur correction function for optically reducing (correcting) an image blur include one that moves a correction lens provided in an interchangeable lens with respect to its optical axis and moves an image sensor provided in a camera respect to the optical axis.

[0003] However, a movable amount of the correction lens is mechanically limited, and a movable amount of the image sensor is limited by, in addition to its mechanical limitation, an image circle formed by the interchangeable lens. Japanese Patent Laid-Open No. 2009-265182 discloses a camera system that changes, depending on movable amounts of a correction lens and an image sensor, a ratio (correction ratio) between image blur correction amounts respectively provided by movements of the correction lens and the image sensor.

[0004] Further, in the lens interchangeable camera system, the center of the image circle may be shifted from the center of the image sensor due to manufacturing errors of the interchangeable lens. Japanese Patent Laid-Open No. 2009-139877 discloses a camera in which an image sensor captures a chart image formed by an interchangeable lens, and moves, using the captured chart image, an initial movement position of the image sensor from its original center position.

[0005] However, the shift amount and the shift direction of the center of the image circle are different in each individual interchangeable lens. As a result, the movable amount of the image sensor when moving within an area of the image circle also differs depending on its moving direction (+ and − directions). For this reason, if the correction ratio is set simply depending on the movable amounts of the correction lens and the image sensor as disclosed in Japanese Patent Laid-Open No. 2009-265182, insufficient image blur correction may be performed. Further, even if the initial movement position of the image sensor is moved from the original center position, the movable amounts of the image sensor in its moving direction are mutually different.

SUMMARY OF THE INVENTION

[0006] The present invention provides a camera and a lens apparatus (an optical apparatus) each capable of effectively utilizing a movable amount of an image sensor even when the center of the image circle is shifted from the center of the image sensor.

[0007] The present invention provides as an aspect thereof a camera used in a camera system including the camera that includes an image sensor movable for image blur correction, and a lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction. The camera includes a setting unit configured to set a ratio of image blur correction amounts provided respectively by movements of the optical element and the image sensor in each of their moving directions, using information on an image circle formed by the lens apparatus, information on a mechanical movable amount of the image sensor, and information on a size of a signal readout area of the image sensor.

[0008] The present invention provides as another aspect thereof a lens apparatus used in a camera system including a camera that includes an image sensor movable for image blur correction, and the lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction. The lens apparatus includes a setting unit configured to set a ratio of image blur correction amounts provided respectively by movements of the optical element and the image sensor in each of their moving directions, using information on an image circle formed by the lens apparatus, information on a mechanical movable amount of the image sensor, and information on a size of a signal readout area of the image sensor.

[0009] The present invention provides as yet another aspect thereof a camera used in a camera system including the camera that includes an image sensor movable for image blur correction, and a lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction. The camera includes a control unit configured to move an initial position of movement of the image sensor to a position at which movable amounts of the image sensor in its moving directions become mutually equal, using information on an image circle formed by the lens apparatus.

[0010] The present invention provides as still another aspect thereof a method of controlling a camera used in a camera system including the camera that includes an image sensor movable for image blur correction, and a lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction. The method includes the step of acquiring information on an image circle formed by the lens apparatus, information on a mechanical movable amount of the image sensor, and information on a size of a signal readout area of the image sensor, and the step of setting a ratio of image blur correction amounts provided respectively by movements of the optical element and the image sensor in each of their moving directions, using the information on the image circle, the information on the mechanical movable amount, and the information on the size of the signal readout area.

[0011] The present invention provides as yet still another aspect thereof a method of controlling a lens apparatus used in a camera system including a camera that includes an image sensor movable for image blur correction, and the lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction. The method includes the step of acquiring information on an image circle formed by the lens apparatus, information on a mechanical movable amount of the image sensor, and information on a size of a signal readout area of the image sensor, and the step of setting a ratio of image blur correction amounts provided respectively by movements of the optical element and the image sensor in each of their moving directions, using the information on the image circle, the information on the mechanical movable amount, and the information on the size of a signal readout area.

[0012] The present invention provides as further another aspect thereof a method of controlling a camera used in a camera system including the camera that includes an image sensor movable for image blur correction, and a lens appa-

ratus that is detachably attachable to the camera and includes an optical element movable for the image blur correction. The method includes the step of acquiring information on an image circle formed by the lens apparatus, and the step of moving the image sensor to a position at which movable amounts of the image sensor in its moving directions become mutually equal, using the information on the image circle.

[0013] The present invention provides as yet further another aspect thereof a computer-readable non-transitory storage medium storing a program for causing a camera or a lens apparatus to execute a process according to any one of the above methods.

[0014] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. **1** is a block diagram illustrating a configuration of a camera system that is Embodiment 1 of the present invention.

[0016] FIG. **2** illustrates an IIS movable amount when the center of an image circle of an interchangeable lens is not shifted with respect to the center of an IIS mechanical stroke range.

[0017] FIG. **3** illustrates an IIS movable amount when the center of the image circle of the interchangeable lens is shifted with respect to the center of the IIS mechanical stroke range.

[0018] FIG. **4** is a flowchart of an image blur correction process performed in Embodiment 1.

[0019] FIG. **5** is a flowchart of an IIS movable amount calculation process performed in Embodiment 1.

[0020] FIG. **6** illustrates a sensor readout area when roll shake correction is performed by IIS in Embodiment 1.

[0021] FIG. **7** illustrates an example of image blur correction performed by movements of a correction lens and an image sensor in Embodiment 1.

[0022] FIG. **8** is a flowchart of an image blur correction process performed in Embodiment 2 of the present invention.

[0023] FIG. **9** is a flowchart of an image blur correction process performed in Embodiment 3 of the present invention.

[0024] FIG. **10** is a flowchart of a sensor effective size setting process performed in Embodiment 3.

[0025] FIG. **11** illustrates a sensor readout area and an EIS clipping area in Embodiment 3.

[0026] FIG. **12** illustrates a sensor effective size in Embodiment 3.

[0027] FIG. **13** illustrates a case where a center position of an image circle is shifted with respect to a center position **1** of an IIS mechanical stroke range in Embodiment 3.

[0028] FIG. **14** is a flowchart of an image blur correction process performed in Embodiment 4 of the present invention.

[0029] FIG. **15** is a flowchart of a process performed in another embodiment of the present invention.

DESCRIPTION OF THE EMBODIMENTS

[0030] Exemplary embodiments of the present invention will hereinafter be described with reference to the accompanying drawings.

Embodiment 1

[0031] FIG. **1** illustrates a configuration of a camera system **10** that is a first embodiment (Embodiment 1) of the present invention. The camera system **10** includes an interchangeable lens **101** as a lens apparatus and a first optical apparatus (one optical apparatus), and a camera body **100** as a camera and a second optical apparatus (another optical device) to which the interchangeable lens **101** is detachably and communicably attached (connected). The camera body **100** includes a camera MPU **102**, an operation unit **103**, an image sensor **104**, camera connection terminals **105**, a camera gyro sensor **106**, and a back display **120**.

[0032] A camera MPU (camera control unit) **102** is a computer that governs overall control of the camera body **100** and the interchangeable lens **101**, and controls various operations such as AE, AF, and image capturing in response to inputs from the operation unit **103**. The camera MPU **102** communicates various commands and information with a lens MPU (lens control unit) **109** as a computer through the camera connection terminals **105** and lens connection terminals **111** provided in the interchangeable lens **101**. The camera connection terminals **105** and the lens connection terminals **111** include a power supply terminal for supplying power from the camera body **100** to the interchangeable lens **101**.

[0033] The operation unit **103** includes a mode dial for setting various image-capturing modes, a release button for instructing start of an image-capturing preparation operation and an image-capturing operation, and other operation members. When the release button is half-pressed, a first switch (SW**1**) is turned on. When the release button is full-pressed, a second switch (SW**2**) is turned on. In response to turning-on of SW**1**, the image-capturing preparation operation (AE and AF) is performed. In response to turning-on of SW**2**, the start of the image-capturing (exposure) operation is instructed, and after a predetermined time from the instruction the image-capturing operation is started. The on/off of SW**1** and SW**2** is notified from the camera MPU **102** to the lens MPU **109** by communication therebetween.

[0034] The image sensor **104** is constituted by a photoelectric conversion element such as a CCD sensor or a CMOS sensor, and photoelectrically converts an object image formed by an image-capturing optical system described later to generate an image-capturing signal. The camera MPU **102** generates a still image or a moving image (video signal) using the image-capturing signal from the image sensor **104**.

[0035] The camera gyro sensor **106** is a shake sensor that detects angular shake of the camera body **100** (hereinafter referred to as "camera shake") due to user's hand jiggling or the like to output a camera shake detection signal as an angular velocity signal. The camera MPU **102** drives an image sensor actuator **107** depending on the camera shake detection signal and on an IIS correction ratio (described later) received from the interchangeable lens **101** to move the image sensor **104** in a direction orthogonal to an optical axis of the image-capturing optical system (or a direction including a directional component orthogonal to the optical axis). The movement of the image sensor **104** reduces (corrects) image blur caused by the camera shake.

[0036] When moving the image sensor **104**, the camera MPU **102** performs feedback control of the image sensor actuator **107** such that a position of the image sensor **104** (a movement amount thereof from a position on the optical

axis that is a movement center) detected by an image sensor position sensor **108** approaches a target position. Thereby, image blur correction (hereinafter referred to as "IIS") by the movement of the image sensor **104** is performed. The IIS is performed in response to camera shake in a vertical (pitch) direction and a horizontal (yaw) direction.

[0037] The rear display **120** constituted by a display device displays a moving image corresponding to the video signal generated by the camera MPU **102** using the image-capturing signal from the image sensor **104**. Before image capturing, the rear display **120** displays a viewfinder image (live-view image) that user can observe. In addition, after image capturing, the rear display **120** can display a still image or a moving image for recording generated by the image capturing. The word "image capturing" in this embodiment means image capturing for recording.

[0038] The interchangeable lens **101** includes the image-capturing optical system (not illustrated, the lens MPU **109** and the lens connection terminal **111***s* described above, and a lens gyro sensor **110**. The lens gyro sensor **110** is a shake sensor that detects angular shake of the interchangeable lens **101** (hereinafter referred to as "a lens shake") to output a lens shake detection signal as an angular velocity signal.

[0039] The lens MPU **109** drives a lens actuator **112** depending on the lens shake detection signal and an OIS correction ratio (described later) to move a correction lens **113** as an optical element that is a part of the image-capturing optical system in a direction orthogonal to an optical axis of the image-capturing optical system. The movement of the correction lens **113** reduces (corrects) image blur caused by the lens shake. When moving the correction lens **113**, the lens MPU **109** performs feedback control of the lens actuator **112** such that a position of the correction lens **113** (a movement amount thereof from a position on the optical axis that is a movement center) detected by a lens position sensor **114** approaches a target position. Thereby, image blur correction by the movement of the correction lens **113** (hereinafter referred to as "OIS") is performed.

[0040] The OIS is performed, as with the IIS, in response to lens shake in the pitch direction and the yaw direction. The correction lens **113** may be moved in the direction orthogonal to the optical axis by being moved in parallel in a plane orthogonal to the optical axis or by being rotated about a point on the optical axis.

[0041] Next, description will be made of a controlled movable amount of the image sensor **104** (hereinafter referred to as "an IIS movable amount") set depending on a relation between a center position of an image circle formed by the interchangeable lens **101** and a center position of a mechanical movable range of the image sensor **104** in the IIS (hereinafter referred to as "an IIS mechanical stroke range"). In the following description, an image blur correctable amount provided by the IIS, which is obtained from the IIS movable amount, is referred to as "an IIS correctable amount". Further, an image blur correctable amount provided by the OIS, which is obtained from a mechanical or controlled movable range of the correction lens **113**, is hereinafter referred to as "an OIS correctable amount".

[0042] In this embodiment, the IIS correctable amount and the OIS correctable amount are expressed as angle (in deg). Further, the above-mentioned IIS correction ratio is a ratio of the IIS correctable amount to a total image blur correctable amount by both the IIS and the OIS, and the OIS correction ratio is a ratio of the OIS correctable amount to the total image blur correctable amount.

[0043] FIG. **2** illustrates the IIS moveable amount in an ideal case where the center position **1'** of the image circle **3** coincides with the center position **1** of the IIS mechanical stroke range **2**. In FIG. **2** and FIG. **3** described later, the right direction is a +X direction, the left direction is a −X direction, the upper direction is a +Y direction, and the lower direction is a −Y direction.

[0044] In the IIS, a signal readout area in an image-capturing surface of the image sensor **104** from which the image-capturing signal for generating display images or recording images (the area is hereinafter referred to as "a sensor readout area") **4** can be moved within the image circle **3** and within the IIS mechanical stroke range **2**. In FIG. **2**, the IIS movable amount **5** in the +X direction and the IIS movable amount **6** in the −X direction are equal to each other. Thus, the IIS correction ratio in the +X direction and the IIS correction ratio in the −X direction are also equal to each other. The same applies to the +Y direction and the −Y direction.

[0045] On the other hand, FIG. **3** illustrates a case where the center position **1'** of the image circle **3** is shifted with respect to the center position **1** of the IIS mechanical stroke range **2** to the lower left due to manufacturing errors or the like of the interchangeable lens **101**. In this case, the IIS movable amount **6'** in the +X direction and the IIS movable amount **5'** in the −X direction are different from each other (specifically, the IIS movable amount **6'** is smaller than the IIS movable amount **5'**). Therefore, the IIS correction ratio in the +X direction and the IIS correction ratio in the −X direction are also different from each other. In such a case, even if the IIS correction ratio is set without considering the shift of the image circle, in other words, moving directions of the image sensor **104**, the IIS correction ratio cannot be realized in one of the +X and the −X directions.

[0046] Specifically, in one of the +X and the −X directions, an IIS correction ratio larger than an actually obtainable IIS correction ratio is set, and thus the sensor movable amount cannot be sufficiently used. Further, due to the above large IIS correction ratio, when the OIS correction ratio smaller than an actually obtainable OIS correction ratio is set, the OIS correctable amount cannot be sufficiently used. As a result, sufficient image blur correction cannot be performed. The same applies to the +Y and −Y directions. In order to solve such a problem, this embodiment performs the following process.

[0047] FIG. **4** illustrates a flowchart of an image blur correction process **1** performed by the camera body **100** (camera MPU **102**) and the interchangeable lens **101** (lens MPU **109**). The process performed by the camera MPU **102** is illustrated on the left side of FIG. **4**, and the process performed by the lens MPU **109** is illustrated on the right side thereof The camera MPU **102** and the lens MPU **109** execute the respective processes according to computer programs stored in the memories **130** and **140** respectively provided in the camera body **100** and the interchangeable lens **101**.

[0048] When the camera body **100** is powered on to supply power to the interchangeable lens **101** and communication between the camera MPU **102** and the lens MPU **109** is started, the camera MPU **102** starts this process at step S**101**. Although the image blur correction process in the X direction will be described below, the same image blur

correction process is performed in the Y direction. The same applies to other processes (flow charts) that will be described later.

[0049] At step S**101**, the camera MPU **102** sends, to the lens MPU **109**, a sensor effective size, a mechanical movable stroke amount of the image sensor **104** in the IIS (hereinafter referred to as "an IIS mechanical stroke amount"), and a roll shake correctable angle in the IIS (hereinafter referred to as "an IIS roll shake correctable angle"). The sensor effective size is a size (a horizontal width and a vertical width) of the above-described sensor readout area **4** in the image sensor **104**. The roll shake correctable angle is a maximum correctable angle in roll shake correction (roll correction) by the IIS for correcting image blur due to the roll shake that is rotational shake around a center of the image sensor **104** (sensor readout area). In this way, the lens MPU **109** acquires (receives) the sensor effective size, the IIS mechanical stroke amount, and the IIS roll shake correctable angle.

[0050] Next at step S**102**, the lens MPU **109** sends, to the camera MPU **102**, information on the image circle of the interchangeable lens **101** (image-capturing optical system), and information on a focal length of the interchangeable lens **101**. The information on the image circle is hereinafter referred to as "image circle information", and the information on the focal length is hereinafter referred to as "focal length information".

[0051] The image circle information includes at least information on an actual center position of the image circle that is shifted from a center position of a designed image circle, that is, the center position of the sensor readout area of the image sensor **104** due to manufacturing errors or the like. In this embodiment, the image circle information further includes information on a radius of the image circle. The image circle information may include information on a diameter of the image circle because the radius can be calculated from the diameter. The image circle information is required when the camera MPU **102** later calculates the IIS movable range. Further, the image circle information may change depending on image-capturing conditions such as a focal length, a focus position and an aperture diameter of the image-capturing optical system, and a posture of the camera system. Therefore, it is desirable that the image circle information sent from the lens MPU **109** to the camera MPU **102** be associated with a value of at least one of parameters of the image-capturing conditions.

[0052] The focal length information is information indicating the focal length of the image-capturing optical system, and is required to convert an image blur correction amount (angle) provided by the movement of the image sensor **104** into a movement amount of the image sensor **104** in control of the IIS.

[0053] At step S**103**, the lens MPU **109** calculates the IIS correctable amount (angle) of the image sensor **104** in each of its moving directions (+X and −X directions). Description will be made of a method of calculating the IIS correctable amount with reference to a flowchart of FIG. **5**

[0054] At step S**201**, the lens MPU **109** calculates the IIS movable amount such that the sensor readout area **4** does not protrude from the image circle **3** in each of the moving directions. When the IIS roll shake correctable angle sent from the camera MPU **102** at step S**101** is set to 0, and thereby the roll shake correction is not performed by the IIS, the lens MPU **109** calculates the IIS movable amount **6'** in the + direction and the IIS movable amount **5'** in the − direction in FIG. **3**.

[0055] When the IIS performs the roll shake correction, the camera MPU **102** sends, to the lens MPU **109**, a roll shake correction angle in the IIS, and the lens MPU **109** calculates the IIS movable amounts **5'**and **6'** in consideration of the roll shake correction angle. FIG. **6** illustrates a range **7** in which the sensor readout area **4** can be moved (rotated) when the IIS performs the roll shake correction. Reference numerals **1**, **1'** to **4** are the same as those in FIG. **3**.

[0056] The range **7** in which the sensor readout area **4** can be rotated is calculated from the sensor effective size and the IIS roll shake correctable angle. Specifically, the IIS movable amount is calculated such that the range **7** in which the sensor readout area **4** can be rotated does not protrude from the image circle **3** in each of the moving directions (+X and −X directions).

[0057] Next, at step S**202**, the lens MPU **109** determines whether or not the IIS movable amount in each of the moving directions calculated from the sensor effective size and the image circle information is smaller than the IIS mechanical stroke amount. If the IIS movable amount is smaller than the IIS mechanical stroke amount, the lens MPU **109** proceeds to step S**203**.

[0058] At step S**203**, the lens MPU **109** sets the IIS movable amount calculated from the sensor effective size and the image circle information to the IIS movable amount used for setting the IIS correction ratio. When the IIS movable amount is lamer than the IIS mechanical stroke amount, the lens MPU **109** proceeds to step S**204** to set a mechanical movable amount, which corresponds to the mechanical stroke amount, to the IIS movable amount to be used for setting the IIS correction ratio.

[0059] Next at step S**104** in FIG. **4**, the lens MPU **109** converts, using the focal length information of the image-capturing optical system, the OIS movable amount into the OIS correctable angle θ_{OIS}. In the following description, the OIS correctable angle θ_{OIS} when the correction lens **113** is moved with respect to the optical axis in the +X direction is referred to as "an OIS+correctable angle θ_{OIS}^{+}", and the OIS correctable angle θ_{OIS} when the correction lens **113** is moved in the −X direction is referred to as "an OIS− correctable angle θ_{OIS}^{-}".

[0060] In addition, at this step, the lens MPU **109** converts, using the focal length information of the image-capturing optical system, the IIS movable amount into the IIS correctable angle θ_{IIS}. In the following description, the IIS correctable angle θ_{IIS} when the image sensor **104** is moved in the +X direction from the initial position (center position of the sensor readout area **4**) is referred to as "an IIS+ correctable angle θ_{IIS}^{+}", and the IIS correctable angle θ_{IIS} when the image sensor **104** is moved in the −X direction is referred to as "an IIS− correctable angle θ_{IIS}^{-}". Since the IIS movable amount differs depending on the moving direction of the image sensor **104**, the IIS correctable angle also differs depending on the moving direction.

[0061] Next at step S**105**, the lens MPU **109** as a setting unit calculates (sets) the OIS correction ratio and the IIS correction ratio in each of the moving directions of the correction lens **113** and the image sensor **104**. The OIS correction ratio indicates a ratio of an image blur correction amount (angle [deg]) provided by the OIS (hereinafter referred to as "an OIS correction amount") to a total image

blur correction amount provided by both the OIS and the IIS. Further, the IIS correction ratio indicates a ratio of an image blur correction amount provided by the IIS (hereinafter referred to as "an IIS correction amount") to the total image blur correction amount.

[0062] The OIS+ correction ratio, the OIS− correction ratio, the IIS+ correction ratio, and the IIS− correction ratio are calculated as follows:

[0063] OIS+ correction ratio: $\theta_{OIS}^{+}/(\theta_{IIS}^{+}+\theta_{OIS}^{+})$

[0064] OIS− correction ratio: $\theta_{OIS}^{-}/(\theta_{IIS}^{-}+\theta_{OIS}^{-})$

[0065] IIS+ correction ratio: $\theta_{IIS}^{+}/(\theta_{IIS}^{+}+\theta_{OIS}^{-})$

[0066] IIS− correction ratio: $\theta_{IIS}^{-}/(\theta_{IIS}^{-}+\theta_{OIS}^{-})$

[0067] Next at step **S106**, the lens MPU **109** sends the IIS+ correction ratio and the IIS− correction ratio calculated at step **S105** to the camera MPU **102**. Then, at step **S107**, the camera MPU **102** controls the IIS using the IIS+ correction ratio and the IIS− correction ratio acquired (received) from the lens MPU **109** at step **S106**. Specifically, the camera MPU **102** calculates the IIS correction amount depending on the camera shake detected by the camera gyro sensor **106** and the IIS± correction ratios. Thereafter the camera MPU **102** controls the image sensor actuator **107** to move the image sensor **104** to a target position corresponding to the IIS correction amount.

[0068] At the same time as step **S107**, at step **S108**, the lens MPU **109** controls the OIS using the OIS+ correction ratio and the OIS− correction ratio calculated at step **S105**. Specifically, the lens MPU **109** calculates the OIS correction amount depending on the lens shake detected by the lens gyro sensor **110** and the OIS± correction ratios. Thereafter, the lens MPU **109** controls the lens actuator **112** to move the correction lens **113** to a target position corresponding to the OIS correction amount.

[0069] Thus, cooperative image blur correction by the OIS and the IIS, that is, by the correction lens **113** and the image sensor **104** is performed.

[0070] Next, with reference to FIG. **7**, description will be made of the cooperative coordinated image blur correction by the OIS and the IIS. The middle part of FIG. **7** illustrates changes in the IIS correction amount by a solid line. Further, the lower part of FIG. **7** illustrates changes in the OIS correction amount by a solid line. Further, the upper part of FIG. **7** illustrates the total image blur correction amount that is the sum of the IIS correction amount illustrated in the middle part of FIG. **7** and the OIS correction amount illustrated in the lower part thereof, and the total image blur correction amount is illustrated by a dotted line in the middle and lower parts. In these drawings, "0" indicates the center position of a mechanical stroke range of the correction lens **113** (hereinafter referred to as "an OIS mechanical stroke range") and the IIS mechanical stroke range.

[0071] In the OIS, the correction lens **113** is moved depending on the lens shake in an OIS+ area and an OIS− area divided at the center position of the OIS mechanical stroke range. On the other hand, in the IIS, the image sensor **104** is moved depending on the camera shake in an IIS+ area and an IIS− area divided at the center position of the IIS mechanical stroke range (initial position).

[0072] As illustrated in the middle of FIG. **7**, the IIS+ correction ratio when the image sensor **104** is moved in the IIS+ area and the IIS− correction ratio when the image sensor **104** is moved in the IIS− area are different from each other. That is, the lens MPU **109** sets, as the IIS+ correction ratio and the IIS− correction ratio, mutually different values depending on the IIS+ correctable angle θ_{IIS}^{+} and the IIS− correctable angle θ_{IIS}^{-}. Since the IIS+ correction ratio and the IIS− correction ratio are different from each other, the OIS+ correction ratio when the correction lens **113** is moved in the OIS+ area and the OIS− correction ratio when the correction lens **113** is moved in the OIS− area are also different from each other.

[0073] As described above, in this embodiment, the IIS+ correction ratio and the IIS− correction ratio are set to appropriate values different from each other depending on the difference between the IIS+ correctable angle and the IIS− correctable angle caused by the shift of the image circle. As a result, the image blur correction can be performed by effectively utilizing the movable amounts of the correction lens **113** and the image sensor **104**.

[0074] In this embodiment, when an optical state such as focus or zoom of the image-capturing optical system changes, the IIS and OIS correctable angles are recalculated. When the camera MPU **102** calculates the IIS and OIS correctable angles, the current optical state of the image-capturing optical system is periodically sent to the camera MPU **102**, and the camera MPU **102** calculates the IIS and OIS correctable angles on the basis of the received optical state and the acquired image circle information. The same applies to the following embodiments.

Embodiment 2

[0075] In a camera system that is a second embodiment (Embodiment 2) of the present invention, the interchangeable lens **101** corresponds to a second optical apparatus and one optical apparatus, and the camera body **100** corresponds to a first optical apparatus and another optical apparatus. In Embodiment 2, the camera body **100** sets, using the OIS correctable angle, the image circle information and the focal distance information received from the interchangeable lens **101**, the OIS correction ratio and the IIS correction ratio in each of the moving directions of the correction lens **113** and the image sensor **104**. The interchangeable lens **101** controls the OIS depending on the OIS correction ratio received from the camera body **100**. The camera system of this embodiment has the same configuration as that of the camera system of Embodiment 1, and the same constituent elements as those in Embodiment 1 are denoted by the same reference numerals as those in Embodiment 1.

[0076] FIG. **8** illustrates a flowchart of an image blur correction process **2** performed by the camera body **100** (camera MPU **102**) and the interchangeable lens **101** (lens MPU **109**). The process performed by the camera MPU **102** is illustrated on the left side of FIG. **8**, and the process performed by the lens MPU **109** is illustrated on the right side thereof. The camera MPU **102** and the lens MPU **109** execute the respective processes according to computer programs.

[0077] When the camera body **100** is powered on to supply power to the interchangeable lens **101** and communication between the camera MPU **102** and the lens MPU **109** is started, the lens MPU **109** starts this process at step **S301**.

[0078] At step **S301**, the lens MPU **109** calculates the OIS correctable angle θ_{OIS} from the OIS movable amount and the focal length information, and sends the calculated OIS correctable angle θ_{OIS} to the camera MPU **102**.

[0079] Next at step **S302**, the lens MPU **109** sends the image circle information and the focal length information to

the camera MPU **102**. The reason therefor is as described at step S**102** in Embodiment 1. Thus, the camera MPU **102** acquires (receives) the image circle information and the focal length information.

[0080] Next at step S**303**, the camera MPU **102** calculates the IIS movable amount in each of the moving directions (+X and −X directions) of the image sensor **104**. The calculation method of the IIS movable amount is the same as that described at step S**103** in Embodiment 1.

[0081] Next at step S**304**, the camera MPU **102** converts, using the focal length information received from the lens MPU **109** at step S**302**, the IIS movable amount calculated at step S**303** into the IIS correctable angle θ_{IIS}.

[0082] Next at step S**305**, the camera MPU **102** as a setting unit calculates (sets) the OIS correction ratio (OIS+ and OIS− correction ratios) and the IIS correction ratio (IIS+ and IIS− correction ratios) in each of the moving directions. The calculation method thereof is the same as that described at step S**105** in Embodiment 1.

[0083] Next at step S**306**, the camera MPU **102** sends the OIS+ and OIS− correction ratios calculated at step S**305**, to the lens MPU **109**.

[0084] Then, at step S**307**, the camera MPU **102** controls the IIS using the IIS+ and IIS− correction ratios calculated at step S**306**, as at step S**107** of the first embodiment.

[0085] At step S**308** simultaneously with step S**307**, the lens MPU **109** controls the OIS using the OIS+ and OIS− correction ratios received from the camera MPU **102** at step S**306** as at step S**108** in Embodiment 1.

[0086] Thus, cooperative image blur correction by the OIS and the IIS, that is, by the correction lens **113** and the image sensor **104** is performed.

[0087] In this embodiment, the lens MPU **109** sends, to the camera MPU **102**, an angle equivalent value as the OIS correctable angle. However, as another embodiment of the present invention, the lens MPU **109** may send, to the camera MPU **102**, a movable amount [mm] of the correction lens **113** instead of the angle equivalent value. When sending the movable amount [mm], the lens MPU **109** separately sends, to the camera MPU **102**, information on image blur correction sensitivity (sensitivity for converting the movement amount [mm] of the correction lens **113** into an angle [deg]). However, it is more desirable that the lens MPU **109** send the OIS correctable angle to the camera MPU **102** because it is possible to reduce a communication amount for the image blur correction sensitivity.

Embodiment 3

[0088] In a camera system that is a third embodiment (Embodiment 3) of the present invention, the interchangeable lens **101** corresponds to a first optical apparatus and one optical apparatus, and the camera body **100** corresponds to a second optical apparatus and another optical apparatus. In Embodiment 3, the camera body **100** performs electronic image blur correction (hereinafter referred to as "EIS") in addition to the IIS. The interchangeable lens **101** sets, using the sensor effective size and the IIS mechanical stroke amount received from the camera body **100**, an OIS correction ratio and an IIS+EIS correction ratio as a total correction ratio of the IIS and EIS performed by the camera body **100**. The camera body **100** controls the IIS and the EIS depending on the IIS+EIS correction ratio received from the interchangeable lens **101**.

[0089] This embodiment is different from the other embodiments in that it is necessary that the lens MPU **109** sets each correction ration in consideration of not only the OIS correctable angle and the IIS correctable angle but also a correctable angle by the EIS (hereinafter referred to as "an EIS correctable angle"). Further, in Embodiment 1, it is necessary that the camera MPU **102** sends the IIS roll shake correctable angle to the lens MPU **109**, and that the lens MPU **109** calculates the IIS movable amount in consideration of the IIS roll shake correctable angle.

[0090] On the other hand, in Embodiment 3, the lens MPU **109** calculates each correction ratio without receiving the EIS correctable angle and the IIS roll shake correctable angle from the camera MPU **102**. This makes it possible to reduce a communication amount between the camera body **100** and the interchangeable lens **101** and to simplify a process of calculating the correction ratio.

[0091] FIG. **9** illustrates a flowchart of an image blur correction process **3** performed by the camera body **100** (camera MPU **102**) and the interchangeable lens **101** (lens MPU **109**). The process performed by the camera MPU **102** is illustrated on the left side of FIG. **9**, and the process performed by the lens MPU **109** is illustrated on the right side thereof The camera MPU **102** and the lens MPU **109** execute the respective processes according to computer programs.

[0092] When the camera body **100** is powered on to supply power to the interchangeable lens **101** and communication between the camera MPU **102** and the lens MPU **109** is started, the camera MPU **102** starts this process at step S**401**.

[0093] At step S**401**, the camera MPU **102** calculates the sensor effective size in consideration of the EIS correctable angle and the IIS roll shake correctable angle. In this calculation, the camera MPU **102** makes the calculated sensor effective size different from the actual size of the sensor readout area **4**. Description will be made of the process of calculating the sensor effective size with reference to a flowchart of FIG. **10**.

[0094] At step S**501**, the camera MPU **102** determines whether or not to perform the EIS in the camera body **100**. When performing the EIS, the camera MPU **102** proceeds to step S**502** to set the sensor effective size to an EIS clipping size that is smaller than the actual size of the sensor readout area **4**. FIG. **11** illustrates the sensor readout area **4** and an EIS clipping area **8** that is an image clipping area in the EIS. The EIS correctable angle is set depending on a maximum amount by which the EIS cropping area **8** can be moved within the sensor readout area **4**. Further, a movable amount of the EIS clipping area **8** within the image circle **3** and within the IIS mechanical stroke range **2** is the sum of a movable amount of the EIS clipping area **8** within the sensor readout area **4** and the IIS movable amount.

[0095] Next at step S**503**, the camera MPU **102** determines whether or not to perform the roll shake correction by the IIS. When the roll shake correction is to be performed, the camera MPU **102** proceeds to step S**504**.

[0096] At step S**504**, the camera MPU **102** sets the sensor effective size that is larger than the actual size of the sensor readout area **4** or than the size of the EIS clipping area **8** when the process at step **502** has been performed.

[0097] FIG. **12** illustrates the sensor effective size **9** in consideration of the roll shake correction amount of the IIS. The sensor effective size **9** is set as a rectangle including

whole of a range **7** in which the sensor readout area **4** is rotated by the roll shake correction. This sensor effective size **9** is expressed by following expression (1). In expression (1), "*" represents multiplication.

$$L_X' = L_X * \cos\theta_{Roll} + L_Y * \sin\theta_{Roll}$$

$$L_Y' = L_X * \sin\theta_{Roll} + L_Y * \cos\theta_{Roll} \quad (1)$$

[0098] In expression (1), Lx' and Ly' respectively represents a horizontal width and a vertical width of the sensor effective size in consideration of the roll shake correction amount of the IIS, and Lx and Ly respectively represents a horizontal width and a vertical width of the sensor effective size that is the same as the sensor readout area **4**. Further. θ_{Roll} represents the IIS roll shake correctable angle.

[0099] At step S**103** in the first embodiment, since the area **7** within which the sensor readout area **4** can be rotated has a complicate shape, calculation of the IIS movable amount in the image circle **3** and in the IIS mechanical stroke range **2** becomes complicated. On the other hand, using the rectangular sensor effective size **9** in consideration of the roll shake correction as in this embodiment makes it possible to simply calculate the IIS movable amount.

[0100] Next at step S**402** in FIG. **9**, the camera MPU **102** sends the sensor effective size calculated at step S**401** and the IIS mechanical stroke amount to the lens MPU **109**. Since the sensor effective size sent at this time has been set in consideration of the EIS correctable angle and the IIS roll shake correctable angle, it is not necessary to send the EIS correctable angle and the IIS roll shake correctable angle to the lens MPU **109**.

[0101] Next at step S**403**, the lens MPU **109** sends the image circle information and the focal length information to the camera MPU **102**. The reason therefor is as described at step S**102** in Embodiment 1. Thus, the camera MPU **102** acquires (receives) the image circle information and the focal length information.

[0102] Next at step S**404**, the lens MPU **109** calculates, using the sensor effective size, the image circle information and the IIS mechanical stroke amount, an I/EIS movable amount in each of the moving directions of the image sensor **104**. The I/EIS means IIS+EIS. The method of calculating the I/EIS movable amount is the same as that described at step S**103** in Embodiment 1. The sensor effective size has been set at step S**401** in consideration of the EIS and the roll shake correction by the IIS, so that the I/EIS movable amount calculated at this step is the sum of the IIS movable amount in consideration of the roll shake correction and the EIS correctable angle (EIS movable amount).

[0103] Next at step S**405**, the lens MPU **109** converts the OIS movable amount and the I/EIS movable amount into correctable angles, respectively, by the same conversion method as described at step S**104** in Embodiment 1.

[0104] At step S**406**, the lens MPU **109** as a setting unit calculates (sets) an OIS correction ratio (OIS+ and OIS− correction ratios) and an I/EIS correction ratio (I/EIS+ and I/EIS− correction ratios) in each of the moving directions. In this embodiment, the OIS correction ratio is a ratio of the OIS correction amount to a total image blur correction amount (angle [deg]) provided by the IIS, the EIS and the OIS, and the I/EIS correction ratio is a ratio of a correction amount (hereinafter referred to as "an I/EIS correction amount") provided by the IIS and the EIS to the above total image blur correction amount.

[0105] Next at step S**407**, the lens MPU **109** sends the I/EIS correction ratio calculated at step S**406** to the camera MPU **102**.

[0106] Then, at step S**408**, the camera MPU **102** calculates the EIS correctable angle and the IIS correctable angle, using the image circle information and the focal length information received from the lens MPU **109** at step S**403**. Further, at step S**407**, the camera MPU **102** divides the I/EIS correction ratio received from the lens MPU **109** depending on a ratio between the EIS correctable angle and the IIS correctable angle. Thus, the camera MPU **102** calculates an IIS correction ratio (IIS+ and IIS− correction ratios) and an EIS correction ratio (EIS+ and EIS− correction ratios) in each of the moving directions of the image sensor **104**. The IIS correction ratio is a ratio of the IIS correction amount to the total image blur correction amount provided by the IIS, the EIS and the OIS. The EIS correction ratio is a ratio of a correction amount (hereinafter referred to as "an EIS correction amount) provided by the EIS to the above total image blur correction amount.

[0107] Next at step S**409**, the camera MPU **102** controls the EIS using the EIS correction ratio (EIS+ and EIS− correction ratios) calculated at step S**408**. Specifically, the camera MPU **102** calculates, depending on the camera shake detected by the camera gyro sensor **106** and the EIS correction ratio, the EIS correction amount that is a movement amount of the EIS clipping area. Then, the camera MPU **102** changes the position of the EIS clipping area depending on the calculated EIS correction amount to output a moving image.

[0108] At step S**410** simultaneously with step S**409**, the camera MPU **102** controls the IIS, using the IIS correction ratio (IIS+ and IIS− correction ratios) calculated at step S**408**, as at step S**107** in Embodiment 1.

[0109] Further, at step S**411** simultaneously with step S**409**, the lens MPU **109** controls the OIS, using the OIS correction ratio (OIS+ and OIS− correction ratios) calculated at step S**407**, as at step S**108** in Embodiment.

[0110] Thus, cooperative image blur correction by the OIS, IIS and EIS is performed.

[0111] In this embodiment, the camera MPU **102** changes the sensor effective size depending on the EIS correction amount and the roll shake correction amount of the IIS to send the changed sensor effective size to the lens MPU **109**. Thus, the lens MPU **109** can calculate (set) the OIS correction ratio and the I/EIS correction ratio without considering the EIS correctable angle and the IIS roll shake correctable angle in the camera body **100**. Therefore, this embodiment can reduce a communication amount between the camera body **100** and the interchangeable lens **101**, and reduce a processing load on the lens MPU **109** when calculating each correction ratio.

Embodiment 4

[0112] In a camera system that is a fourth embodiment (Embodiment 4) of the present invention, the interchangeable lens **101** corresponds to a first optical apparatus and one optical apparatus, and the camera body **100** corresponds to a second optical apparatus and another optical apparatus. In Embodiment 4, the camera body **100** changes the initial position (hereinafter referred to as "an IIS initial position") of the movement of the image sensor **10**, which is different from the other embodiments, in order to eliminate the necessity of calculating the IIS correction ratio in each of the

moving directions of the image sensor **104**. The interchangeable lens **101** calculates the OIS correction ratio and the IIS correction ratio, using the IIS movable amount received from the camera body **100**. Then, the camera body **100** controls the IIS depending on the IIS correction ratio received from the interchangeable lens **101**.

[0113] With reference to FIG. **13**, description will be made of the change of the IIS initial position. In the left and right parts of FIG. **13**, as in FIG. **3**, the case is illustrated where the center position **1'** of the image circle **3** is shifted to the lower left with respect to the center position **1** of the IIS mechanical stroke range **2**. In the left part of FIG. **13**, a case is illustrated where the IIS movable amount **6'** in the +X direction and the IIS movable amount **5'** in the −X direction when the IIS initial position is located at the center position **1** of the IIS mechanical stroke range **2**. The IIS movable amounts **5'** and **6'** are mutually different. Similarly, the IIS movable amount **12'** in the +Y direction and the IIS movable amount **11'** in the −Y direction are mutually different. Therefore, it is necessary to calculate and set the IIS correction ratio in each of the moving directions (+X, −X, +Y and −Y directions) of the image sensor **104** as in the other embodiments.

[0114] On the other hand, in the right part of FIG. **13**, a case is illustrated where the IIS initial position is set to a position at which the same movable amount can be obtained in the + and − directions in each of the X and Y directions. In this case, the IIS movable amount **6"** in the +X direction and the IIS movable amount **5"** in the −X direction are equal to each other, and the IIS movable amount **12"** in the +Y direction and the IIS movable amount **11"** in the −Y direction are equal to each other. Therefore, it is not necessary to calculate the IIS correction ratio in each of the moving directions of the image sensor **104**. That is, the same IIS correction ratio can be used in the +X direction and the −X direction, and the same IIS correction ratio can be used in the +Y direction and the −Y direction.

[0115] FIG. **14** illustrates a flowchart of an image blur correction process **4** performed by the camera body **100** (camera MPU **102**) and the interchangeable lens **101** (lens MPU **109**). The process performed by the camera MPU **102** is illustrated on the left side of FIG. **14**, and the process performed by the lens MPU **109** is illustrated on the right side thereof. The camera MPU **102** and the lens MPU **109** execute respective processes according to computer programs.

[0116] When the camera body **100** is powered on to supply power to the interchangeable lens **101** and communication between the camera MPU **102** and the lens MPU **109** is started, the lens MPU **109** starts this process at step S**601**.

[0117] At step S**601**, the lens MPU **109** sends the image circle information to the camera MPU **102**. Thus, the camera MPU **102** acquires (receives) the image circle information.

[0118] Next at step S**602**, the camera MPU **102** calculates, using the sensor effective size, the image circle information and the IIS mechanical stroke amount, the IIS movable amount in each of the moving directions of the image sensor **104**. The method of calculating the IIS movable amount is the same as that described at step S**103** in Embodiment 1.

[0119] Next at step S**603**, the camera MPU **102** as a control unit moves the IIS initial position (that is, moves the image sensor **104**) to a position at which the IIS movable amounts in the respective moving directions (+X and −X directions) calculated at step S**602** become equal to each other. Then, the camera MPU **102** recalculates the IIS movable amount. In this calculation, since the IIS movable amounts in the respective moving directions are equal to each other, it is only necessary to calculate the IIS movable amount in one moving direction. The IIS movable amount after the movement of the IIS initial position is expressed by following expression (2).

$$d_x' = (d_x^+ + d_x^-)/2 \quad (2)$$

[0120] In expression (2), d_x' represents the IIS movable amount in the +X direction and −X direction after the movement of the IIS initial position, d_x^+ represents the IIS movable amount in the +X direction, and d_x^- is the IIS movable amount in the −X direction.

[0121] Next at step S**604**, the camera MPU **102** sends the IIS movable amount calculated at step S**603** to the lens MPU **109**. As described above, since the IIS movable amounts in the respective moving directions are equal to each other, it is only necessary to send the IIS movable amount in one moving direction to the lens MPU **109**.

[0122] Next at step S**605**, the lens MPU **109** sends the focal length information of the image-capturing optical system to the camera MPU **102**.

[0123] Next at step S**606**, the lens MPU **109** converts, as at step S**104** in Embodiment 1, the OIS movable amount and the IIS movable amount respectively into the OIS correctable angle and the IIS correctable angle.

[0124] Next at step S**607**, the lens MPU **109** as a setting unit calculates (sets), from a ratio of the OIS correctable angle and the IIS correctable angle calculated at step S**606**, the OIS correction ratio and the IIS correction ratio. In this calculation, since the IIS correctable angle is common to the +X and −X directions, it is not necessary to calculate the IIS correction ratios in these directions.

[0125] Next at step S**608**, the lens MPU **109** sends the common IIS correction ratio in the +X and −X directions to the camera MPU **102**. Then, at step S**609**, the camera MPU **102** controls the IIS, using the IIS correction ratio received from the lens MPU **109** at step S**606**, as at step S**107** in Embodiment 1.

[0126] At step S**610** simultaneously with step S**609**, the lens MPU **109** controls the OIS control, using the OIS correction ratio calculated at step S**607**, as at step S**108** in Embodiment 1.

[0127] Thus, cooperative image blur correction by both the OIS and the IIS, that is, by the correction lens **113** and the image sensor **104** is performed.

[0128] In this embodiment, the IIS initial position is moved to the position at which the IIS movable amounts in the respective moving directions (+X and −X directions) of the image sensor **104** are equal to each other, so that the IIS correction ratios in the respective moving directions become equal to each other. This makes it possible to eliminate the necessity of calculating the IIS correction ratios in the respective moving directions. Therefore, it is not necessary to communicate the IIS correction ratios in the respective moving directions between the camera body **100** and the interchangeable lens **101**, thereby reducing the communication amount therebetween. Further, it is not necessary to calculate the IIS correction ratios in the respective moving directions, so that the process load of the lens MPU **109** can be reduced.

[0129] Embodiment 1 described the case where the camera MPU **102** sends the sensor effective size, the IIS

mechanical stroke amount and the IIS roll shake correctable angle to the lens MPU **109**, and the lens MPU **109** calculates the IIS movable amount in each of the moving directions of the image sensor **104**. However, the camera MPU **102** may calculate the IIS movable amount in each of the moving directions in advance using the image circle information and the focal length information to send them to the lens MPU **109**, and the lens MPU **109** may calculate the IIS correction ratio depending thereon.

[0130] Further, Embodiment 3 described the case where the lens MPU **109** calculates the I/EIS movable amount and the I/EIS correction ratio. However, the camera MPU **102** may calculate the I/EIS movable amount and the I/EIS correction ratio. In this case, the camera MPU **102** may convert the sensor effective size into the sensor effective size in consideration of the roll shake correction amount to reduce the process load of the camera MPU **102** when calculating the I/EIS movable amount and the I/EIS correction ratio.

[0131] Furthermore, a process may be performed in which the image blur correction process **1** described in Embodiment 1 is performed when the camera body **100** performs still image capturing, and the image blur correction processing **2** described in Embodiment 2 is performed when the camera body **100** performs moving image capturing. FIG. **15** illustrates this process performed by the camera MPU **102**.

[0132] At step S**701**, the camera MPU **102** determines whether the camera body **100** performs still image capturing or moving image capturing. When performing the still image capturing, the camera MPU **102** proceeds to step S**702** to perform the image blur correction process **1** with the lens MPU **109**. On the other hand, when performing the moving image capturing, the camera MPU **102** proceeds to step S**703** to perform the image blur correction process **2** with the lens MPU **109**.

[0133] In this case, an appropriate image blur correction process can be selectively performed in the still image capturing and in the moving image capturing.

Other Embodiments

[0134] Embodiments of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiments and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiments, and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiments and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiments. The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0135] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0136] This application claims the benefit of Japanese Patent Application No. 2018-167529, filed on Sep. 7, 2018 which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. A camera used in a camera system including the camera that includes an image sensor movable for image blur correction, and a lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction, the camera comprising:

a setting unit configured to set a ratio of image blur correction amounts provided respectively by movements of the optical element and the image sensor in each of their moving directions, using information on an image circle formed by the lens apparatus, information on a mechanical movable amount of the image sensor, and information on a size of a signal readout area of the image sensor.

2. The camera according to claim **1**, wherein the setting unit is configured to set the ratio depending on an optical image blur correctable amount provided by the movement of the optical element in each of the moving directions of the optical element, and on an image blur correctable amount provided by the movement of the image sensor in each of the moving directions of the image sensor.

3. The camera according to claim **1** wherein the information on the image circle is information on a center position of the image circle and one of a radius and a diameter of the image circle.

4. The camera according to claim **3**, wherein the setting unit is configured to set the ratio in each of the moving directions when the center position of the image circle is shifted from a center position of the signal readout area.

5. The camera according to claim **2**, wherein the setting unit is configured to set the ratio, using smaller one of the image blur correctable amount of the image sensor and the mechanical movable amount of the image sensor in each of the moving directions of the image sensor as the image blur correctable amount of the image sensor to be used for setting the ratio.

6. The camera according to claim **1**, wherein the setting unit is configured to send the ratio to the lens apparatus.

7. The camera according to claim **1**, wherein the setting unit is configured to set the ratio, when the image sensor is rotated for image blur correction against roll shake, using a larger size of the signal readout area of the image sensor than an actual size of the signal readout area as the size of a signal readout area of the image sensor to be used for setting the ratio.

8. The camera according to claim **1**, wherein the setting unit is configured to set the ratio, when electronic image blur correction is performed on the image sensor, using a smaller

size of the signal readout area of the image sensor than an actual size of the signal readout area as the size of a signal readout area of the image sensor to be used for setting the ratio.

9. A lens apparatus used in a camera system including a camera that includes an image sensor movable for image blur correction, and the lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction, the lens apparatus comprising:

a setting unit configured to set a ratio of image blur correction amounts provided respectively by movements of the optical element and the image sensor in each of their moving directions, using information on an image circle formed by the lens apparatus, information on a mechanical movable amount of the image sensor, and information on a size of a signal readout area of the image sensor.

10. The lens apparatus according to claim **9**, wherein the setting unit is configured to set the ratio depending on an optical image blur correctable amount provided by the movement of the optical element in each of the moving directions of the optical element, and on a sensor image blur correctable amount provided by the movement of the image sensor in each of the moving directions of the image sensor.

11. The lens apparatus according to claim **9**, wherein the information on the image circle is information on a center position of the image circle and one of a radius and a diameter of the image circle.

12. The lens apparatus according to claim **11**, wherein the setting unit is configured to set the ratio in each of the moving directions when the center position of the image circle is shifted from a center position of the signal readout area.

13. The lens apparatus according to claim **10**, wherein the setting unit is configured to set the ratio, using smaller one of the sensor image blur correctable amount and the mechanical movable amount of the image sensor in each of the moving directions of the image sensor as the sensor image blur correctable amount to be used for setting the ratio.

14. The lens apparatus according to claim **9**, wherein the setting unit is configured to send the ratio to the camera.

15. The lens apparatus according to claim **9**, wherein the setting unit is configured to set the ratio, when the image sensor is rotated for image blur correction against roll shake, using a larger size of the signal readout area of the image sensor than an actual size of the signal readout area as the size of a signal readout area of the image sensor to be used for setting the ratio.

16. The lens apparatus according to claim **9**, wherein the setting unit is configured to set the ratio, when electrical image blur correction is performed on the image sensor, using a smaller size of the signal readout area of the image sensor than an actual size of the signal readout area as the size of a signal readout area of the image sensor to be used for setting the ratio.

17. A camera used in a camera system including the camera that includes an image sensor movable for image blur correction, and a lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction, the camera comprising:

a control unit configured to move an initial position of movement of the image sensor to a position at which movable amounts of the image sensor in the moving directions of the image sensor become mutually equal, using information on an image circle formed by the lens apparatus.

18. The camera according to claim **17**, further comprising a setting unit configured to set a ratio of image blur correction amounts provided respectively by movements of the optical element and the image sensor, the ratio being common to moving directions of each of the optical element and the image sensor.

19. A method of controlling a camera used in a camera system including the camera that includes an image sensor movable for image blur correction, and a lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction, the method comprising the steps of:

acquiring information on an image circle formed by the lens apparatus, information on a mechanical movable amount of the image sensor, and information on a size of a signal readout area of the image sensor; and

setting a ratio of image blur correction amounts provided respectively by movements of the optical element and the image sensor in each of their moving directions, using the information on the image circle, the information on the mechanical movable amount, and the information on the size of the signal readout area.

20. A method of controlling a lens apparatus used in a camera system including a camera that includes an image sensor movable for image blur correction, and the lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction, the method comprising the steps of:

acquiring information on an image circle formed by the lens apparatus, information on a mechanical movable amount of the image sensor, and information on a size of a signal readout area of the image sensor; and

setting a ratio of image blur correction amounts provided respectively by movements of the optical element and the image sensor in each of their moving directions, using the information on the image circle, the information on the mechanical movable amount, and the information on the size of a signal readout area.

21. A method of controlling a camera used in a camera system including the camera that includes an image sensor movable for image blur correction, and a lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction, the method comprising the steps of:

acquiring information on an image circle formed by the lens apparatus; and

moving the image sensor to a position at which movable amounts of the image sensor in the moving directions of the image sensor become mutually equal, using the information on the image circle.

22. A computer-readable non-transitory storage medium storing a program for causing a camera, which is used in a camera system including the camera that includes an image sensor movable for image blur correction, and a lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction, to execute a process according to the method of claim **19**.

23. A computer-readable non-transitory storage medium storing a program for causing a lens apparatus, which is used in a camera system including a camera that includes an image sensor movable for image blur correction, and the lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction, to execute a process according to the method of claim **20**.

24. A computer-readable non-transitory storage medium storing a program for causing a camera, which is used in a camera system including the camera that includes an image sensor movable for image blur correction, and a lens apparatus that is detachably attachable to the camera and includes an optical element movable for the image blur correction, to execute a process according to the method of claim **21**.

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